JINGTAO LEI

Applied Mathematics Undergraduate | AI & Machine Learning Research

Central South University · Dundee International Institute

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ACADEMIC BACKGROUND

Bachelor of Science (B.Sc.) in Applied Mathematics

Central South University, Dundee International Institute — Changsha, China

Sep 2024 – Jun 2028 (Expected) | Cumulative Average: 87.74/100

ACADEMIC & RESEARCH EXPERIENCE

Position	Organization / Department	Supervisor	Period
SURP Research Student	CUHK, Computer Science & Engineering	Prof. Shengchao Liu	Jun 2026 – Present
Undergraduate Researcher	Multimodal Learning Research Project	Prof. David P. Woodruff	Jan–Jul 2026

SELECTED RESEARCH ACTIVITIES

Video Diffusion Models toward World-Model Capabilities

The Chinese University of Hong Kong — Department of Computer Science and Engineering

Supervisor: Prof. Shengchao Liu | Jun 2026 – Present

- Investigated whether pretrained **video diffusion models encode usable physical dynamics**, with the broader goal of extending visual generative models toward **world-model capabilities** for prediction and physical reasoning.
- Built controlled motion and two-object interaction benchmarks with known physical ground truth and a reproducible **CogVideoX-2B** analysis pipeline; mapped direction, speed, optical flow, and contact information across Transformer layers, diffusion noise levels, and readouts using linear probes, ID/OOD evaluation, and negative controls.
- Found strong linear decodability of motion factors: best controlled ID results reached approximately **1.09° direction error** and **R^2 = 0.988** for speed. The factor map varied across layers, noise levels, and readouts, motivating stronger tests of functional use.
- Designed and initiated **representation-level interventions** using factor-subspace removal, matched random orthogonal controls, and temporal perturbations to distinguish information that is merely decodable from information that is functionally used by the model.

Sample-Wise Dynamic Prompt Truncation for Multimodal Learning

Undergraduate Research Project

Supervisor: Prof. David P. Woodruff | Jan 2026 – Jul 2026 | Code: GitHub Repository

- Investigated whether **prompt length can be treated as an adaptive resource** in parameter-efficient multimodal learning rather than assigning every input the same fixed prompt budget.
- Extended a ViT-based **Mixture of Prompt Experts (MoPE)** classifier with a lightweight layer-wise gating mechanism conditioned on visual and textual representations to dynamically determine the retained visual prompt length.
- Adapted and evaluated the framework on a derived **IU X-Ray chest X-ray classification** task, covering multimodal data processing, model reproduction, architecture modification, and controlled fixed-versus-adaptive prompting experiments.
- In the recorded experiment, dynamic truncation reduced the effective visual prompt budget from **six tokens to one** while maintaining validation performance; best validation accuracy increased from **0.8969 to 0.8996**, and final validation accuracy from **0.8802 to 0.8963**.

AREAS OF EXPERTISE

Multimodal learning, video diffusion, world models, representation learning, adaptive computation

TECHNICAL SKILLS

Programming & Scientific Computing: Python, Bash, NumPy, pandas, Matplotlib

Machine Learning: PyTorch, Hugging Face Transformers & Diffusers, Vision Transformers, multimodal learning, prompt tuning, diffusion and video diffusion models

Research Methods: controlled dataset construction, representation probing, intermediate-feature analysis, ID/OOD evaluation, negative controls, ablation studies, representation-level interventions

Systems & Research Computing: Linux, SSH-based remote development, Git, Slurm, Singularity, GPU/HPC workflows; hands-on experience running experiments on NVIDIA H200 clusters

MANUSCRIPT IN PREPARATION

Jingtao Lei, Hongji Li, Dexiang Shu.
Sample-Wise Dynamic Prompt Truncation in a MoPE-Style Multimodal Classifier for Chest X-Ray Classification.
Manuscript in preparation, 2026. | Code: GitHub Repository

AWARDS & ACADEMIC ACHIEVEMENTS

- **First Prize**, 21st Central South University Mathematical Modeling Competition and Internal Selection for the National Competition
- **Third Prize**, Annual Comprehensive Student Evaluation Award, Central South University
- **97.80/100 (A+)**, **From Combinatorics to Calculus**, taught by Prof. Dan Ciubotaru, University of Oxford

ADDITIONAL TRAINING

AWS Academy Graduate — Cloud Foundations